# Top chemistry generator model by language and mode

Google Patents · highest average full grade (total_score, out of 40) · 10 generator models · Gemini 3.8 Flash grading

| | Assertion verification | Comparison | Conceptual | Constraint search | Lookup | Process retrieval | Source finding |
|---|---|---|---|---|---|---|---|
| **English** | **5-model tie**<br>Claude Haiku / DeepSeek / GPT-5.6 Luna / Qwen3.8-27B / GLM-5.3<br>Avg 32.00 / 40 · n=7/6/6/8/6 | **2-model tie**<br>DeepSeek / Qwen3.8-27B<br>Avg 32.00 / 40 · n=5/5 | **Claude Haiku**<br>Avg 31.75 / 40 · n=8 | **Ternary Bonsai**<br>Avg 31.80 / 40 · n=5 | **Qwen3.8-27B**<br>Avg 31.56 / 40 · n=9 | **DeepSeek**<br>Avg 31.50 / 40 · n=6 | **2-model tie**<br>Claude Haiku / Qwen3.8-27B<br>Avg 32.00 / 40 · n=6/7 |
| **French** | **Qwen3.8-27B**<br>Avg 32.00 / 40 · n=9 | **DeepSeek**<br>Avg 31.79 / 40 · n=14 | **Claude Haiku**<br>Avg 31.43 / 40 · n=7 | **Qwen3.8-27B**<br>Avg 31.86 / 40 · n=7 | **Qwen3.8-27B**<br>Avg 32.20 / 40 · n=10 | **Qwen3.8-27B**<br>Avg 31.78 / 40 · n=9 | **Claude Haiku**<br>Avg 32.38 / 40 · n=8 |
| **Spanish** | **2-model tie**<br>DeepSeek / Qwen3.8-27B<br>Avg 32.00 / 40 · n=8/6 | **Qwen3.8-27B**<br>Avg 32.00 / 40 · n=3 | **2-model tie**<br>DeepSeek / GLM-5.3<br>Avg 32.00 / 40 · n=6/6 | **Ternary Bonsai**<br>Avg 29.62 / 40 · n=8 | **2-model tie**<br>Claude Haiku / DeepSeek<br>Avg 32.00 / 40 · n=2/3 | **3-model tie**<br>Claude Haiku / DeepSeek / Ternary Bonsai<br>Avg 32.00 / 40 · n=3/3/3 | **5-model tie**<br>Claude Haiku / DeepSeek / GPT-5.6 Luna / Ternary Bonsai / Qwen3.8-27B<br>Avg 32.00 / 40 · n=4/4/4/5/4 |
| **German** | **3-model tie**<br>Claude Haiku / DeepSeek / GLM-5.3<br>Avg 32.00 / 40 · n=2/2/2 | **Claude Haiku**<br>Avg 31.67 / 40 · n=3 | No graded samples | **Qwen3.8-27B**<br>Avg 31.00 / 40 · n=2 | **6-model tie**<br>Claude Haiku / DeepSeek / GPT-5.6 Luna / Ternary Bonsai / Qwen3.8-27B / GLM-5.3<br>Avg 32.00 / 40 · n=2/2/2/2/2/1 | **GPT-5.6 Luna**<br>Avg 31.67 / 40 · n=3 | **Solar Pro4**<br>Avg 33.67 / 40 · n=3 |
| **Chinese** | **3-model tie**<br>Claude Haiku / DeepSeek / Qwen3.8-27B<br>Avg 32.00 / 40 · n=5/5/4 | **Qwen3.8-27B**<br>Avg 32.25 / 40 · n=4 | **Qwen3.8-27B**<br>Avg 32.14 / 40 · n=7 | **GLM-5.3**<br>Avg 30.71 / 40 · n=7 | **Claude Haiku**<br>Avg 31.40 / 40 · n=5 | **GLM-5.3**<br>Avg 32.00 / 40 · n=3 | **3-model tie**<br>Claude Haiku / Qwen3.8-27B / GLM-5.3<br>Avg 32.00 / 40 · n=7/7/7 |

Blue = one winner · amber = exact tie · gray = no graded samples. Ties are determined before rounding scores.
n = graded candidate rows per winner; tie counts follow the displayed model order. All 1,950 graded outputs are included.
Some winning averages use only 1–3 outputs; sample sizes differ across models and language–mode pairs.
Short labels: Claude Haiku 5.5 · DeepSeek v4.1 Flash · GLM-5.3 FlashX · Ternary Bonsai 2 27B. Full model IDs are in the CSV.